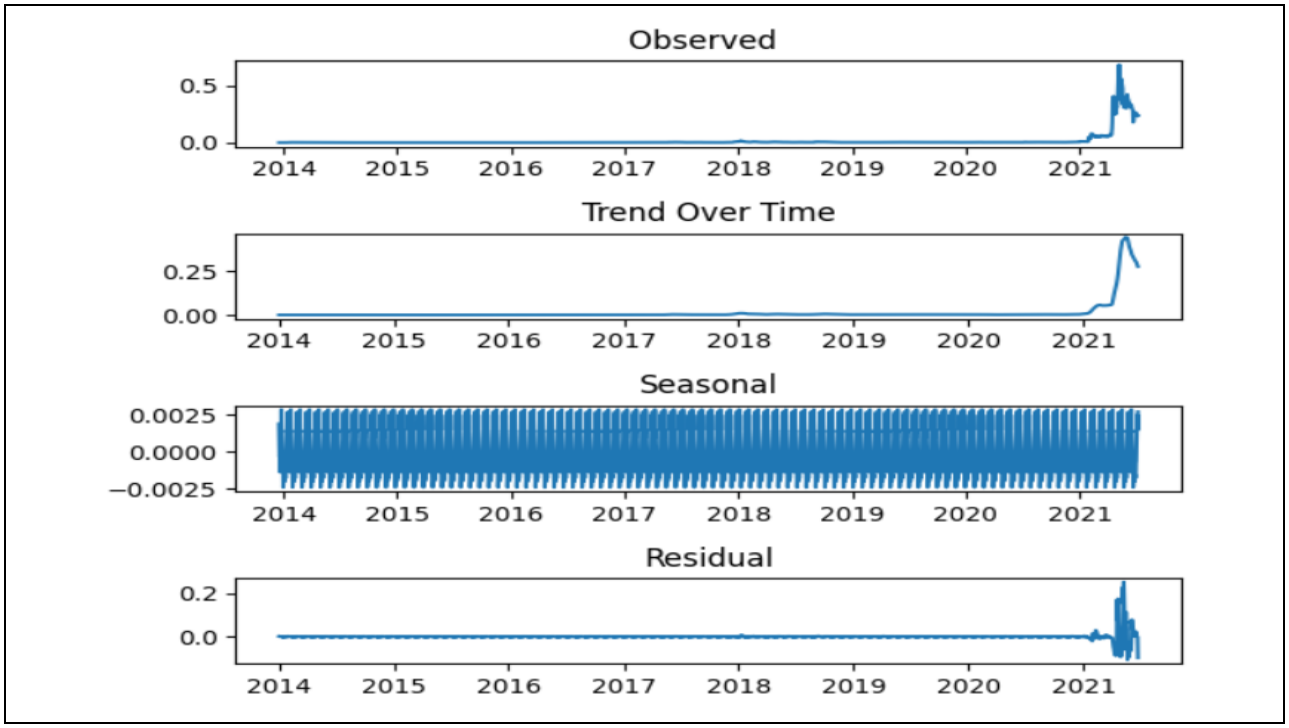
Time Series Analysis

Financial Data Analysis in Python

DCS520

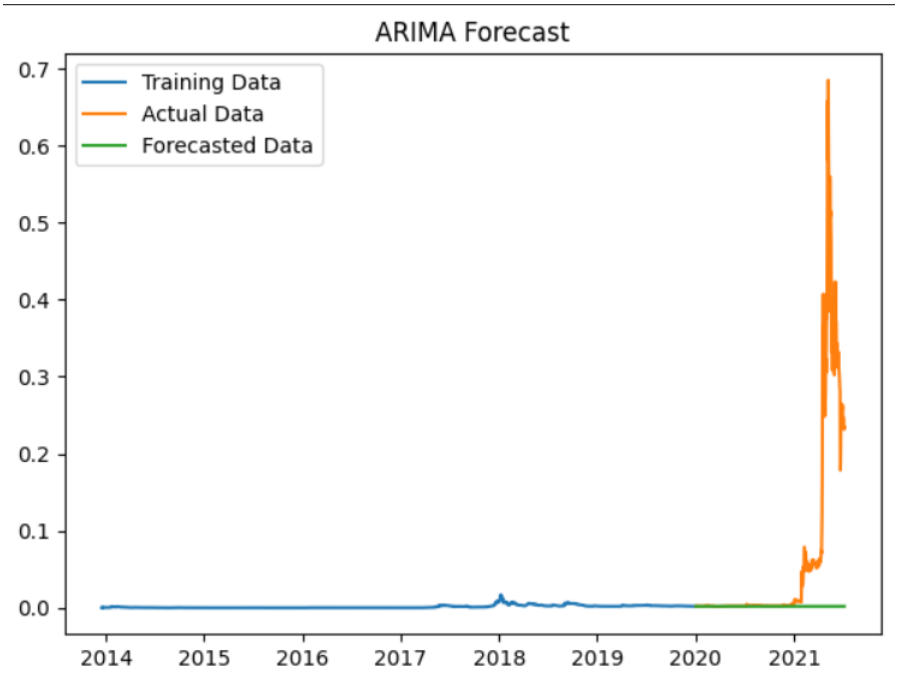
Overview: In week 5 of Financial Data Analysis in Python mini project, the chosen dataset containing historical price of cryptocurrency Dogecoin including Price Statistics (High, Low, Open, Close), Volume and Market Cap has been analyzed. Time Series Analysis and Forecasting Models such as ARIMA(Auto Regression Integrated Moving Average) has been applied to the Price(Close) and Date.

Time Series Decomposition: Time-series decomposition is a technique used to break down a time series into its constituent components such as trends, seasonality, and residuals. This helps in understanding the underlying patterns in the data.



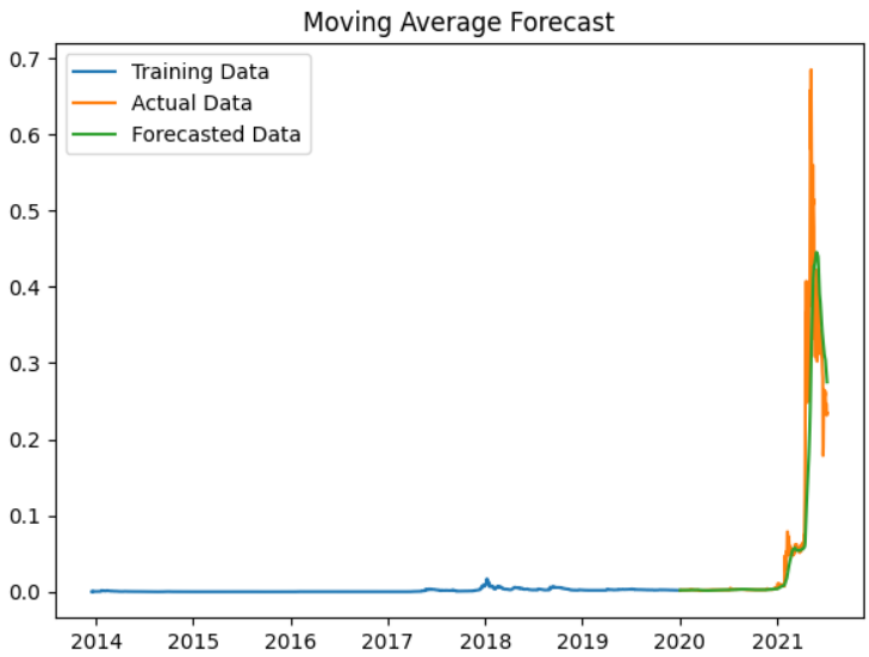
- **Methodology**: Additive decomposition was used, assuming that the components add up to the observed data. A period of 30 days was chosen to capture monthly seasonality.
- **Results**: The “Trend Over Time” component shows a general up and down movement in Dogecoin prices over time. The seasonal component reveals recurring patterns, such as periodic price fluctuations. The residuals represent noise or irregular fluctuations that cannot be explained by trend or seasonality.

Forecasting Model (ARIMA): ARIMA (AutoRegressive Integrated Moving Average) is a widely used statistical model for time-series forecasting. It combines three components as AutoRegressive which Captures the relationship between an observation and its lagged values. Integrated which differences the data to make it stationary and Moving Average which captures the relationship between an observation and residual errors from a moving average model.



- **Implementation**: The data was split into training (80%) and testing (20%) sets. An ARIMA model with order (5, 1, 0) was fitted to the training data. Forecasts were generated for the test set. The model's performance was evaluated using MSE, which measures the average squared difference between actual and predicted values.

Moving Average Model: A moving average model is a simple yet effective forecasting technique that smooths out short-term fluctuations and highlights longer-term trends.



- **Implementation**: A 30-day moving average was calculated for the training data. The moving average was used to forecast the test set. The model's performance was evaluated using Mean Squared Error.

Conclusion: The report shows a comprehensive workflow for analyzing and forecasting Dogecoin prices using time series techniques. Major steps including preprocessing the data, decomposing it into trend, seasonality, and residuals, and evaluating the performance of ARIMA and moving average models. While ARIMA provided better accuracy, the moving average model offered simplicity and ease of interpretation.